

Experiments: Part II

PS200C Quant Methods in Politics, Spring 2026

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Threats to Internal and External Validity for Experiments

- **Internal validity**: can we estimate the treatment effect for our particular sample?
 - Fails when there are differences between treated and controls (other than the treatment itself) that affect the outcome
 - In other words, how might the randomization have failed? (See next)

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 - Fails when there are differences between treated and controls (other than the treatment itself) that affect the outcome
 - In other words, how might the randomization have failed? (See next)
- **External validity**: can we extrapolate our estimates to other populations?
 - Might Y_{1i} , Y_{0i} , and τ_i have looked different in a different part of the population?

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- Differential attrition

- attrition *can* be okay, even if different rates for treated and controls
- but, if probability of attrition depends on treatment status **and** potential outcomes, you break the randomization
- Two scenarios: (i) people attrite from treatment-taking, but you still get to measure outcome; (ii) people attrite and you can't measure any outcome for them post-treatment.

Example: Clingingsmith, Khwaja, Kremer



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- Wealthier Pakistanis tend to use private Hajj tour operators rather than the lottery.
- “Parties” apply to go, and are randomly selected, with stratification on sect, region, and accommodation.
- Compliance with the experiment is imperfect:
 - 99% who win lottery attend the Hajj.
 - 11% who lose lottery still attend the Hajj (via private tours).

Example: Natural Experiment in Pakistan

Two kinds of information to bolster the randomization assumption:

1 Qualitative information:

- The lottery selection algorithm was designed and implemented by an independent and reputable third party, and there were no reports of lottery manipulation.

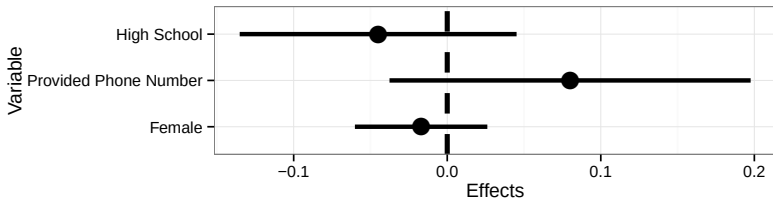
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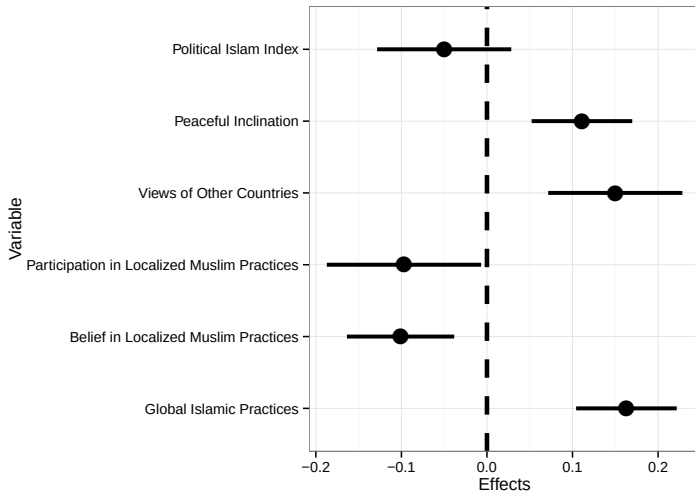
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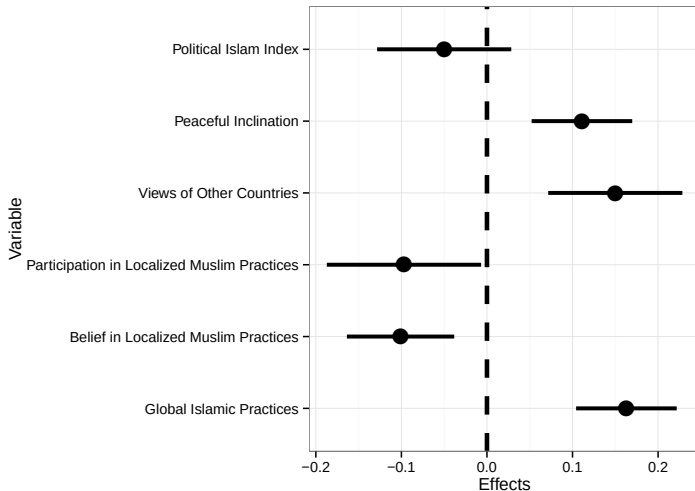
2 Quantitative information: Balance tests on *observables*



Difference in Means: lottery winner vs. losers



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Would you say these are the “effects of going on the Hajj”?

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- actual treatments may be bundled

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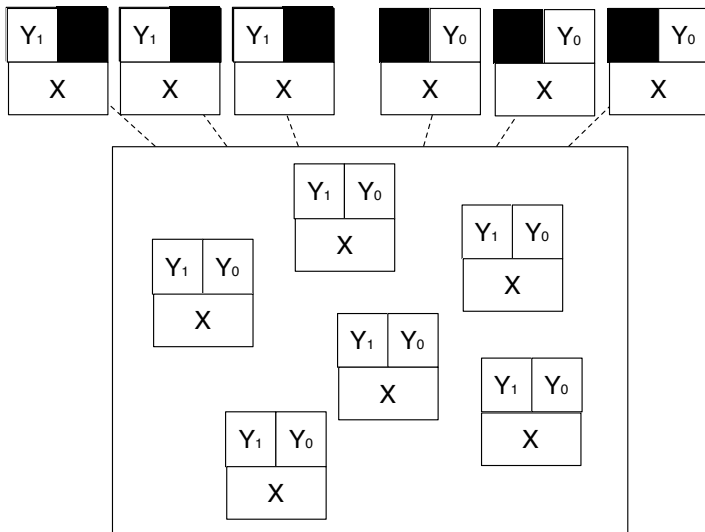
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- Our aim is to get away from “valid” and “not valid” to instead discuss the evidence that a given conclusion would hold.

Covariates and Experiments



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- Good idea to (i) check “balance”, and/or (ii) adjust for chance imbalances to at least improve efficiency.

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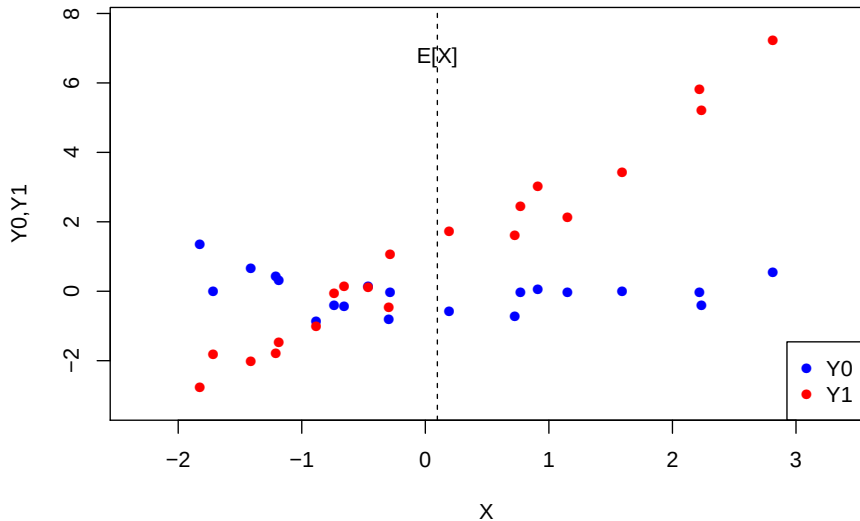
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- But, do not control for **post-treatment** covariates!
- This is an over-simplification: sometimes post-tx variables are okay, sometimes your QOI involves post-tx control, and sometimes pre-tx variables are problematic...we will need DAGs for that.

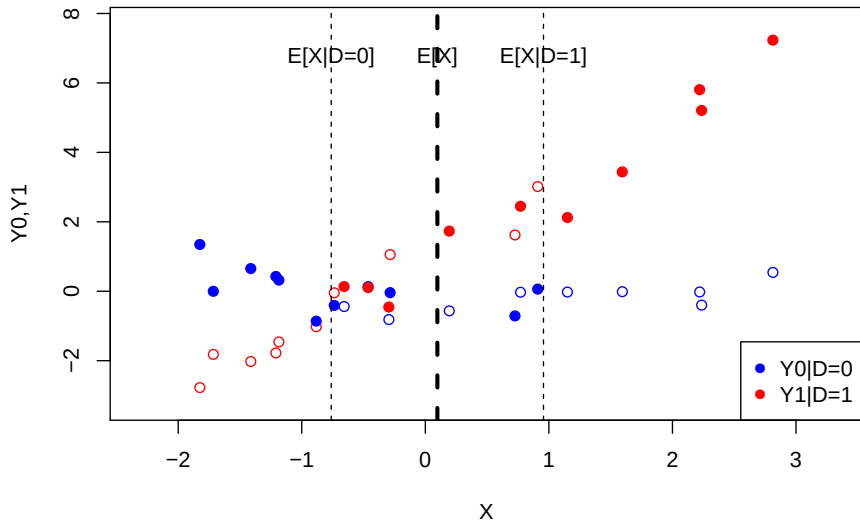
Regression Adjustment for Experiments

True ATE = 1.169



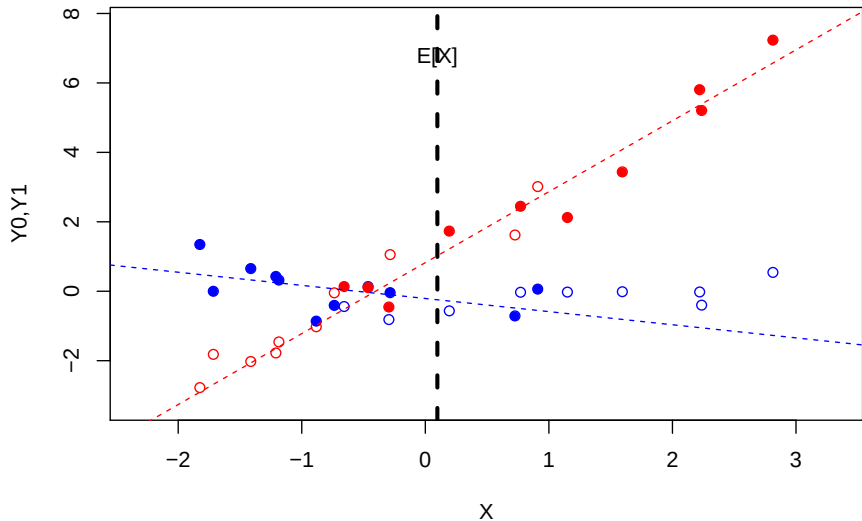
Regression Adjustment For Experiments

Est. ATE = 2.698



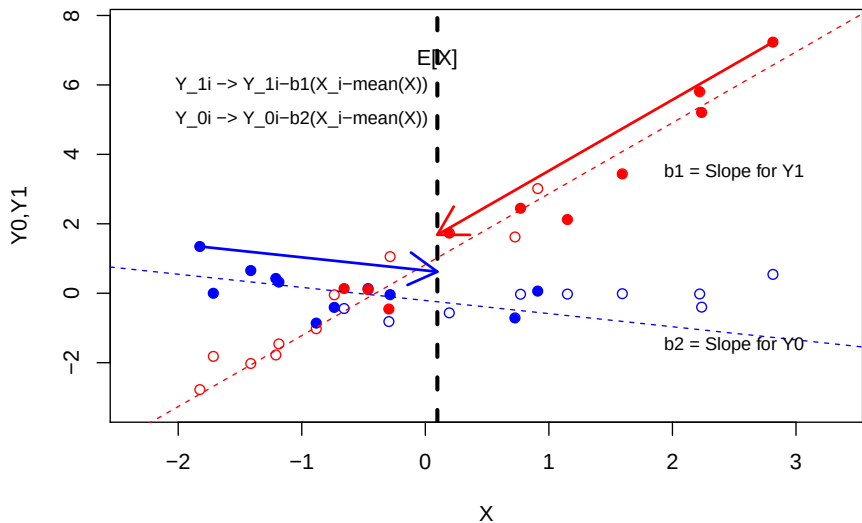
Regression Adjustment For Experiments

Consider regression lines (interacting with D)



Regression Adjustment For Experiments

Regression Adjusted ATE = 1.102



Covariate Adjustment with Regression

Freedman (2008) noted that regression of the form:

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Better still, (see this working paper):

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- this is also equivalent to some other methods you may hear about (g-computation, t-learner, regression imputation)

Why the weird demeaning business in this model?

It is isomorphic to a model without the demeaning, but makes interpretation for ATE more immediate.

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So,

- τ_{Lin} tells you comparison after “moving” all points to $X_i = \bar{X}$
- model could be used to give you $ATE(X)$ but conveniently, $\mathbb{E}_x[ATE(X)] = \tau_{Lin}$.

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- What is $\mathbb{E}[Y_i | D_i = 1, X_i = x^*] - \mathbb{E}[Y_i | D_i = 0, X_i = x^*]$ for $x^* \neq \bar{X}$?

So,

- τ_{Lin} tells you comparison after “moving” all points to $X_i = \bar{X}$
- model could be used to give you $ATE(X)$ but conveniently, $\mathbb{E}_x[ATE(X)] = \tau_{Lin}$.
- allowing this interaction is a good idea, emerging as standard practice

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- Helpful both to protect against chance imbalance and improve efficiency.
- Since covariates are balanced by design, results typically not model dependent (see appendix for FWL explanation)
- Best reason to be skeptical of adjustment: fishing.
 - always report unadjusted result first
 - best if adjustment strategy pre-specified

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 - Simple random assignment into two treated/control will place $\{2, 2\}$ and $\{8, 8\}$ together in the same group one-third of the time

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or more generally, if there are J strata or blocks, then

$$\hat{\tau}_B = \sum_{j=1}^J \frac{N_j}{N} \hat{\tau}_j$$

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If (empirical) $Pr(D = 1)$ changes by block, OLS differently weights the blocks.

We can correct for this by going back to the Lin-style (interacted) regression!

- Again see this working paper
- Another proposal is to weight each observation so that block-treatment probabilities become the same. Let

$$p_{ij} \equiv Pr(D_i = 1 \mid \text{block for unit } i)$$

$$w_{ij} = \left(\frac{1}{p_{ij}} \right) D_i + \left(\frac{1}{1 - p_{ij}} \right) (1 - D_i)$$

...which turns out to be exactly the same thing as the interactive regression

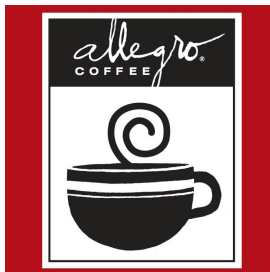
Fair Trade Labeling Experiment (Hainmueller et al, 2012)

Label Experiment

Treatment



Control



The map displays the locations of FT Labels (red dots) and Control Labels (blue dots) across the Northeast United States. The main map shows Massachusetts (MA) with labels for Andover, Bedford, Woburn, Framingham, Wayland, Newton, Wellesley, Brighton, Cambridge, Melford, Boston, and Hingham. Three inset maps show: Maine (ME) with Portland; Rhode Island (RI) with Providence and Cranston; and Connecticut (CT) with Hadley, Glastonbury, and West Hartford. A legend indicates that red dots represent FT Labels and blue dots represent Control Labels.

Example: Fair Trade Labeling Experiment

```
> d <- read.dta("FTdata.dta")  
> head(d)
```

	store	pair	FTweek	lnsalesd
1	1	1	1	3.20
2	4	1	0	2.77
3	6	2	1	4.18
4	9	2	0	4.04
5	21	3	1	4.30
6	24	3	0	3.93

Example: Fair Trade Labeling Experiment

```
> cr.out <- lm(lnsalesd~FTweek,data=d)
> coeftest(cr.out,vcov = vcovHC(cr.out, type = "HC1"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.35000	0.16079	27.0537	<2e-16 ***
FTweek	0.12385	0.21424	0.5781	0.5686

--

Example: Fair Trade Labeling Experiment

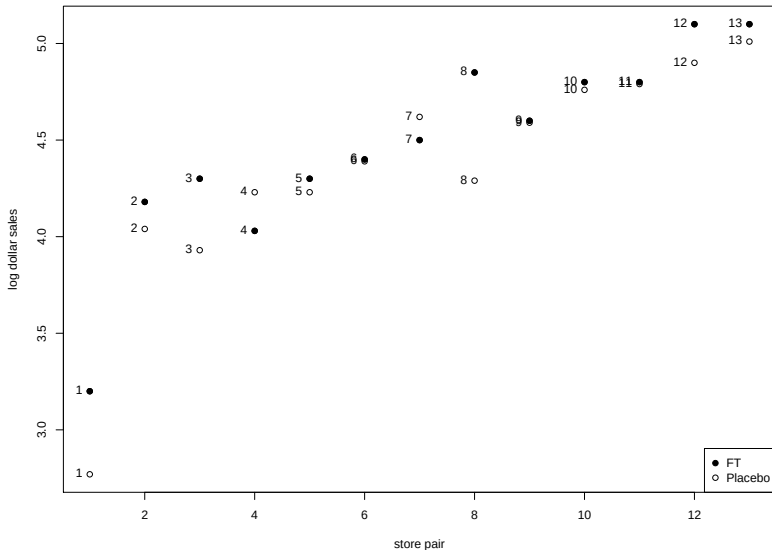
```
> br.out <- lm(lnsalesd~FTweek+as.factor(pair),data=d)
> coeftest(br.out,vcov = vcovHC(br.out, type = "HC1"))
```

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.923077	0.162144	18.0277	4.671e-10	***
FTweek	0.123846	0.060176	2.0581	0.0619840	
as.factor(pair)2	1.125000	0.159549	7.0511	1.335e-05	***
as.factor(pair)3	1.130000	0.204440	5.5273	0.0001304	***
as.factor(pair)4	1.145000	0.231925	4.9369	0.0003439	***
as.factor(pair)5	1.280000	0.161773	7.9123	4.208e-06	***
as.factor(pair)6	1.410000	0.169987	8.2948	2.591e-06	***
as.factor(pair)7	1.575000	0.203689	7.7324	5.317e-06	***
as.factor(pair)8	1.585000	0.277319	5.7154	9.675e-05	***
as.factor(pair)9	1.610000	0.169987	9.4713	6.420e-07	***
as.factor(pair)10	1.795000	0.165195	10.8660	1.450e-07	***
as.factor(pair)11	1.810000	0.169987	10.6479	1.810e-07	***
as.factor(pair)12	2.015000	0.164183	12.2729	3.763e-08	***
as.factor(pair)13	2.070000	0.160298	12.9134	2.127e-08	***

--

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Conclusions: Blocking

How does blocking help?

- Increases efficiency if the blocking variables predict outcomes (see appendix for the variance argument)
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- Two constraints: data you can get, number of units to work with.
- Irrelevant variables: burn up DOFs
- Most important: predictors of potential outcomes!
- Pre-Tx outcomes is great; add other main predictors if available/helpful
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How to block?

- Stratification
- Pair-wise matching; two pair-matching

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- If doing randomization inference, block design influences construction of Ω : re-randomization is within blocks.
- Also, type of SE can influence type of bootstrap, e.g. use the block bootstrap (resample at level of blocks) for clustered SEs

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- Finding partners, manage relationships will be common failure point
- Iron law of field experiments: keep it simple. Many ways to fail.
- Getting difficult to publish studies that only examine the efficacy of some intervention: look for substantively interesting treatment and outcome.

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- Possible tradeoff between internal validity and external validity

Appendix

Frisch-Waugh-Lovell (FWL) Theorem

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All together, this gives the Frisch-Waugh-Lovell (FWL) theorem:

$$\beta_k = \frac{\text{Cov}(\tilde{Y}_i, \tilde{X}_{ki})}{\text{Var}(\tilde{X}_{ki})}$$

Where,

- $\tilde{X}_{ki}$ is the residual from a regression of X_{ki} on all other covariates
- $\tilde{Y}_i$ may have all other X partialled out as well or you can use the original Y_i .

Why are Experimental Findings Robust to Alternative Specifications?

If our “ X_k ” of interest is the D :

$$\hat{\beta}_D = \frac{\widehat{\text{Cov}}(\tilde{Y}_i, \tilde{D}_i)}{\widehat{\text{Var}}(\tilde{D}_i)}$$

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- For experimental data, on average, what will $\tilde{D}_i$ look like?
- What does this suggest about the effect of adding covariates to the simple regression of Y on D ?

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Imagine models for complete and block-randomized designs:

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where R_D^2 is R^2 from regression of D on all block indicators and a constant.

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$$\begin{aligned} \text{Var}[\hat{\tau}_{CR}] &= \frac{\sigma_{\epsilon}^2}{\sum_{i=1}^n (D_i - \bar{D})^2} & \text{with } \hat{\sigma}_{\epsilon}^2 &= \frac{\sum_{i=1}^n \hat{\epsilon}_i^2}{n-2} = \frac{SSR_{\hat{\epsilon}}}{n-2} \\ \text{Var}[\hat{\tau}_{BR}] &= \frac{\sigma_{\epsilon^*}^2}{\sum_{i=1}^n (D_i - \bar{D})^2 (1 - R_j^2)} & \text{with } \hat{\sigma}_{\epsilon^*}^2 &= \frac{\sum_{i=1}^n \hat{\epsilon}_i^{*2}}{n-k-1} = \frac{SSR_{\hat{\epsilon}^*}}{n-k-1} \end{aligned}$$

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where R_D^2 is R^2 from regression of D on all B variables and a constant.

- If R_D^2 is high, variance inflated. But what can we say about R_D^2 when we're talking about blocking?
- If blocks explain outcomes strongly, how will $\hat{\sigma}_{\epsilon^*}^2$ compare to $\hat{\sigma}_{\epsilon}^2$?

When does blocking in the analysis help?

$$\begin{aligned} \text{Var}[\hat{\tau}_{CR}] &= \frac{\sigma_{\varepsilon}^2}{\sum_{i=1}^n (D_i - \bar{D})^2} & \text{with } \hat{\sigma}_{\varepsilon}^2 &= \frac{\sum_{i=1}^n \hat{\varepsilon}_i^2}{n-2} = \frac{SSR_{\hat{\varepsilon}}}{n-2} \\ \text{Var}[\hat{\tau}_{BR}] &= \frac{\sigma_{\varepsilon^*}^2}{\sum_{i=1}^n (D_i - \bar{D})^2 (1 - R_D^2)} & \text{with } \hat{\sigma}_{\varepsilon^*}^2 &= \frac{\sum_{i=1}^n \hat{\varepsilon}_i^{*2}}{n-k-1} = \frac{SSR_{\hat{\varepsilon}^*}}{n-k-1} \end{aligned}$$

where R_D^2 is R^2 from regression of D on all B variables and a constant.

- If R_D^2 is high, variance inflated. But what can we say about R_D^2 when we're talking about blocking?
- If blocks explain outcomes strongly, how will $\hat{\sigma}_{\varepsilon^*}^2$ compare to $\hat{\sigma}_{\varepsilon}^2$?
- Adding DOF adjustment, we end up asking whether $\frac{SSR_{\hat{\varepsilon}^*}}{n-k-1} < \frac{SSR_{\hat{\varepsilon}}}{n-2}$

How much is explained by blocks? (Fair Trade Example)

```
> summary(lm(lnsalesd~as.factor(pair),data=d))
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.9850	0.1212	24.621	2.72e-12	***
as.factor(pair)2	1.1250	0.1715	6.562	1.82e-05	***
as.factor(pair)3	1.1300	0.1715	6.591	1.74e-05	***
as.factor(pair)4	1.1450	0.1715	6.678	1.52e-05	***
as.factor(pair)5	1.2800	0.1715	7.466	4.73e-06	***
as.factor(pair)6	1.4100	0.1715	8.224	1.65e-06	***
as.factor(pair)7	1.5750	0.1715	9.186	4.77e-07	***
as.factor(pair)8	1.5850	0.1715	9.245	4.44e-07	***
as.factor(pair)9	1.6100	0.1715	9.390	3.71e-07	***
as.factor(pair)10	1.7950	0.1715	10.469	1.05e-07	***
as.factor(pair)11	1.8100	0.1715	10.557	9.56e-08	***
as.factor(pair)12	2.0150	0.1715	11.752	2.68e-08	***
as.factor(pair)13	2.0700	0.1715	12.073	1.94e-08	***

--

Residual standard error: 0.1715 on 13 degrees of freedom

Multiple R-squared: 0.9474, Adjusted R-squared: 0.8988

F-statistic: 19.5 on 12 and 13 DF, p-value: 2.356e-06